

BMW Sales – Power BI Project Documentation

Project Description

The BMW Car Sales Dashboard was developed to analyze and visualize sales performance across multiple dimensions such as model, year, region, fuel type, color, transmission, and engine size. The dataset, sourced from Kaggle, provided comprehensive sales information that enabled detailed exploration of trends and patterns.

The project was built entirely in **Power BI**, using **Power Query** for data cleaning and transformation to maintain data accuracy and consistency. Dynamic measures and key performance indicators (KPIs) were created using **DAX**, allowing the dashboard to update automatically when users filter by attributes like year or model. The interactive design helps users quickly identify high-performing models and regions while understanding how different vehicle characteristics influence sales performance.

Business Problems and Analytical Objectives

The goal of this dashboard was to provide a structured and interactive view of BMW's sales data to support decision-making in marketing, operations, and strategy. The analysis focused on answering the following key questions:

- Which BMW models and regions contribute most to overall sales performance?
- How do factors such as fuel type, transmission, and color influence sales distribution?
- What patterns can be observed in sales trends over time, and how do different models perform across multiple years?
- How can insights from the data help identify opportunities for improving product strategy, market focus, and inventory planning?

Each visualization in the dashboard was designed with a specific objective in mind, helping to compare performance across categories, highlight sales variations, and reveal the relationships between product attributes and sales outcomes.

Outcome and Insights

The BMW Car Sales Dashboard provides a clear and interactive understanding of sales performance from different perspectives. Users can explore the data dynamically, applying filters for attributes such as year, model, or region to uncover meaningful insights without needing additional analysis.

This project demonstrates the ability to combine **data cleaning**, **data analysis**, and **data visualization** to deliver a complete business intelligence solution. The integration of **Power Query**, **DAX**, and advanced **Power BI visual techniques**, including drill-downs, hierarchies, and calculated KPIs, ensures that the dashboard remains flexible, insightful, and user-friendly.

By integrating data preparation, analytics, and visualization, the project showcases a full analytical workflow that supports informed, data-driven decision-making for business stakeholders.